

1.16 Solution: The total energy, without the extra conductor, is given by

$$E_i[\psi_i] = \frac{1}{2} \int_V \nabla \psi_i \cdot \nabla \psi_i d^3x,$$

where ψ_i is the solution to the initial configuration as in Problem 1.15, and V is the volume exterior to the conductors. Now, introduce the extra conductor, which occupies a volume of V' , the total energy in the final configuration, assuming no change to the potential, is

$$E_f[\psi_i] = \frac{1}{2} \int_{V \setminus V'} \nabla \psi_i \cdot \nabla \psi_i d^3x,$$

i.e., the total energy calculated with the initial potential excluding the volume taken by the extra conductor. Obviously,

$$E_i[\psi_i] > E_f[\psi_i].$$

However, $E_f[\psi_i]$ is not the correct total energy of the final configuration. From variational principle, we know that, for the correct final potential ψ_f , the energy is a minimum for the final configuration, and any other potential can only lead to a higher energy, *i.e.*,

$$E_f[\psi_i] > E_f[\psi_f],$$

which means

$$E_f[\psi_i] > E_f[\psi_i] > E_f[\psi_f].$$

That is to say, the introduction of an extra conductor will lower the total energy of the system.